Chapter 9

Lesson 3-12-2025

9.1 Channel estimation

3 Dec 2025

General idea (pilots) and formal writing of $\bar{Y}$, vec$(\bar{Y})$

Ref: ***A signal processing perspective (pp. 57-130), p. 54***

We've seen OFDM and single carrier equalization. Today professor wants to finish channel estimation. If we want to transmit with OFDM, we need to estimate the channel.

$$\mathsf{y}[k] = \sqrt{G}\mathsf{h}[k]\mathsf{p}[k] + \mathsf{v}[k]$$

↑ *pilot component*

$$\xrightarrow{\text{IDFT}} \text{of length } K$$

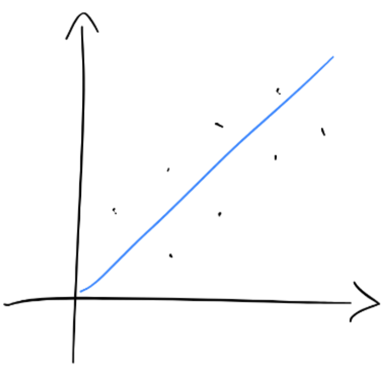
How can we do channel estimation? There are two approaches:

- Maximum likelihood
- Least squares $\longrightarrow$ ***like MD (Minimum Distance)***

In the book the difference between least square and MMSE is not made clear.

If we have something like this and we're asked to draw a line which minimizes the error we're doing, that is Least Squares (LS).

How do we do channel estimation? We use what we'll call a **pilot**: this is a symbol at a certain frequency and time that we know.



LS minimization visualization

Since we know the symbol we've sent and we know its power, we can learn the response of the channel accordingly.

Formal writing:

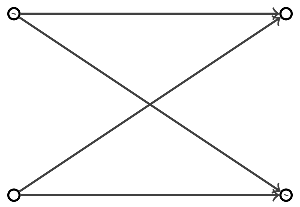
$$\mathsf{y}[k] = \underset{\substack{\uparrow\\ \text{useful}}}{\sqrt{G}\mathsf{h}[k]\mathsf{p}[k]} + \underset{\substack{\uparrow\\ \text{noise}}}{\mathsf{v}[k]}$$

We can model the received signal as

$$y[n] = \sqrt{G}\sum_{\ell=0}^{L} H[\ell]x[n-\ell] + v[n].$$

How the H matrices represent the delay

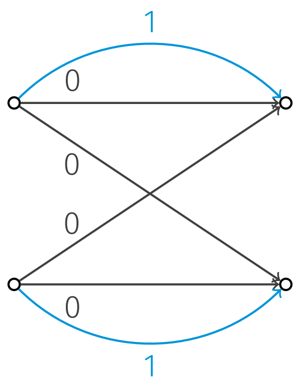
Let's suppose we have 2 transmit and 2 receive antennas and all the signals arrive without any delay at the receiver:



We'll use a matrix such as

$$H = \begin{bmatrix} h_{00} & h_{01} \\ h_{10} & h_{11} \end{bmatrix}$$

to characterize the channel. Even in this case, there could be other paths, with different delays!



Here the channel won't be flat anymore (the arrows in blue arrive at time 1): the one before will be

$$H(0) = \begin{bmatrix} h_{00}(0) & h_{01}(0) \\ h_{10}(0) & h_{11}(0) \end{bmatrix}$$

and we'd also have

$$H(1) = \begin{bmatrix} h_{00}(1) & 0 \\ 0 & h_{11}(1) \end{bmatrix};$$

we have some form of reflection which makes the signal arrive later at the receiver.

Of course if we were to have reflections between 0,1 and 1,0, we'd have

$$H(1) = \begin{bmatrix} h_{00}(1) & h_{01}(1) \\ h_{10}(1) & h_{11}(1) \end{bmatrix}.$$

It's similar to what we had in the SISO case:

$$y[n] = \sum_{\ell=0}^{L} h[\ell]x[n-\ell]; \tag{9.1}$$

this is why we wrote the channel as

$$y[n] = \sqrt{G}\sum_{\ell=0}^{L} H[\ell]x[n-\ell] + v[n]. \tag{9.2}$$

Previous symbols interfere with current symbols!

This is the IO relationship. If we want to estimate the channel, what do we want to do? We don't transmit an unknown symbol x, but rather some **known symbols**; this is indicated with p, which stands for ***pilots***.

Before diving further, let's address the values which they'll use. We are indeed sending N_p pilots, but since the channel has L taps, all received symbols up until the $L-1$th will have **contributions which we don't want.** In order to remove such contributions **we only consider the received signals from L to $N_p - 1$.**

Specifically, we have the synthesized form

$$\begin{aligned} &[y[L] \quad \dots \quad y[N_p-1]] = \\ &= \sqrt{G}[H[0] \quad H[1] \quad \dots \quad H[L]] \begin{bmatrix} p[L] & p[L+1] & \dots & p[N_p-1] \\ p[L-1] & p[L] & \dots & p[N_p-2] \\ \vdots & \ddots & \ddots & \vdots \\ p[0] & p[1] & \dots & p[N_p-L-1] \end{bmatrix} + \bar{V} \\ &\leftrightarrow \bar{Y} = \sqrt{G}\bar{H}\bar{P} + \bar{V}. \end{aligned} \tag{9.3}$$

Where:

- since each pilot is a $N_t \times 1$ vector, the $\bar{P}$ matrix will be a $N_t(L+1) \times (N_p - L)$ matrix.
- Since each $H[\ell]$ matrix is a $N_r \times N_t$ matrix, the $\bar{H}$ matrix will be a $N_r \times N_t(L+1)$ matrix.
- Both $\bar{Y}, \bar{V}$ will be $N_r \times (N_p - L)$ as a result of the product $\bar{H}\bar{P}$.

N_p is the number of pilots we transmit and it's up to us to decide its value. If we transmit too many pilots, we can't transmit the information symbols. We need to transmit some pilots (generally a vector of all pilots) and then some pilots scattered around the time and frequency domains.

The book, in order to make the notation more compact, uses the Kronecker product of two matrices.

> Of course in order to actually understand this, we need to explode this notation.

We also have the vec operator. We have a specific property we'll use:

Relation with the Kronecker product

We have that

$$\text{vec}(ABC) = (C^T \otimes A)\text{vec}(B), \tag{9.4}$$

$$\text{vec} : \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a \\ c \\ b \\ d \end{bmatrix}$$

where $\otimes$ represents the Kronecker product.

Even for SISO, we used to make a compact notation by using matrices which imply time dependance.

> This is similar to how with OFDM we use a matrix for representing the transform

> If we look at exercise 20, we'll end up with 2 matrices representing the channel

After going to our matrix notation, we further progress to a vector notation, knowing that if we were to stack one on the other all the values of $\bar{Y}$, we'd get

$$\begin{aligned} \text{vec}(\bar{Y}) &= \sqrt{G}\,\text{vec}(\bar{H}\bar{P}) + \text{vec}(\bar{V}) \\ &= \sqrt{G}(\bar{P}^T \otimes I_{N_r})\text{vec}(\bar{H}) + \text{vec}(\bar{V}) \\ &= \sqrt{G}P_{\otimes}\,\text{vec}(H) + \text{vec}(\bar{V}). \end{aligned} \tag{2.229}$$

We can apply the forementioned property; in applying it, we use $A = I_{N_r}$, we can develop like it's written above here. Question is: what's $P_{\otimes}$? It's just $\bar{P}^T \otimes I_{N_t}$.

9.2 Linear estimation for a MIMO AWGN channel

Now we estimate, either with ML or LS.

In this estimation, **we assume that we know the conditional distribution of the received signal given the channel.** In other words, we know that our transmit receive relation will follow (2.229) and that our noise $\bar{V}$ will be AWGN. Thanks to the fact that we can choose our pilots to be IID, we know that the **ML estimator will be equivalent to the MAP estimator.**

Considering (2.229), we have that $P_\otimes$ is known and that the observation given by a certain channel realization h (which realizes the distribution of vec($\bar{H}$)) is complex Gaussian with mean

$$\mathbb{E}[\text{vec}(\bar{Y})] = \mathbb{E}[\sqrt{G}\bar{P}_\otimes \underbrace{\mathsf{h}}_{\text{realization}} + \text{vec}(\bar{V})] = \sqrt{G}\bar{P}_\otimes \mathsf{h} + 0 \tag{9.5}$$

and covariance

$$\begin{aligned}\mathbb{E}[(\text{vec}(\bar{Y}) - \mu_{\text{vec}(\bar{Y})})(\text{vec}(\bar{Y}) - \mu_{\text{vec}(\bar{Y})})^*] &= \mathbb{E}[\text{vec}(\bar{V})\text{vec}^*(\bar{V})] \\ &= N_0 I. \end{aligned}\tag{9.6}$$

We'll thus have that, given a certain realization of the channel, vec($\bar{Y}$) is Gaussian with PDF (according to *Uni Generale/Digital Communications/Complex Gaussian distribution > 57a95e*)

$$f_{\text{vec}(\bar{Y})|\text{vec}(\bar{H})}(y|\mathsf{h}) = \frac{1}{(\pi N_0)^{(N_p - L)N_r}} \exp\left(-\frac{\|y - \sqrt{G}\bar{P}_\otimes \mathsf{h}\|^2}{N_0}\right), \tag{2.230}$$

which, thanks to the monotonicity of the exponential, is equivalent to minimizing $-\log f_{\text{vec}(\bar{Y})|\text{vec}(\bar{H})}(\cdot)$, ultimately giving

$$\text{vec}(\hat{\bar{H}})(\mathsf{y}) = \arg\min_{\mathsf{h}} \|y - \sqrt{G}\bar{P}_\otimes \mathsf{h}\|^2. \tag{2.232}$$

This is identical to the deduction of the zero forcing equalizer we already discussed (equations (2.145), (2.146)), ultimately giving

$$\text{vec}(\hat{\bar{H}})(\mathsf{y}) = \underbrace{\frac{1}{\sqrt{G}}(\bar{P}_\otimes^* \bar{P}_\otimes)^{-1}\bar{P}_\otimes^*}_{W^{\text{LS}*}} y, \tag{2.233}$$

and thus the linear **least squares estimator**

$$W^{\text{LS}} = \frac{1}{\sqrt{G}}\bar{P}_\otimes(\bar{P}_\otimes^* \bar{P}_\otimes)^{-1} \tag{2.233+}$$

to the observation (we have $^{-1}$ and not † since we know that $\bar{P}_\otimes^* \bar{P}_\otimes$ is a square matrix).

How does this tie together with the ZF filters?
For ZF, we were trying to minimize

$$\|\sqrt{G}\bar{w}_j^*\bar{T} - \bar{d}_j^*\|^2. \tag{2.138-ish}$$

and we got to

$$\bar{w}_j^{\text{ZF}} = \frac{1}{\sqrt{G}}(\bar{T}^*)^\dagger \bar{d}_j \quad \text{time domain} \tag{2.145}$$
$$\bar{W}^{\text{ZF}} = \frac{1}{\sqrt{G}}(\bar{T}^*)^\dagger \bar{D}. \quad \text{z domain} \tag{2.146}$$

we know that the Moore-Penrose pseudoinverse for a tall matrix A is

$$A^\dagger = (A^*A)^{-1}A^*$$

if we then consider the complex conjugate of the argument of (2.232), we get

$$\|y^* - \sqrt{G}\mathsf{h}^*\bar{P}_\otimes^*\|,$$

which fits neatly in (2.138-ish), to give (from 2.145)

$$\mathsf{h} = \frac{1}{\sqrt{G}}(\bar{P}_\otimes)^\dagger y,$$

which finally leads to (2.233).

> Professor would like of us to remember the reasoning and that we introduced $\otimes$ and vec only for the reasons of having vectors, since with vectors we know the joint probability distribution of a Gaussian vector.

Sorry... which type of estimator are we talking about?
We have that this estimator is both the ML, MAP and LS estimator!
Note though that this is NOT the LMMSE estimator!!

Due to the discussion we made about zero forcing equalizers, **we need to ensure that $\bar{P}_\otimes$ is square or tall**, requiring to check that

$$N_p \geq (L+1)N_t + L \tag{2.234}$$

and also that $\bar{P}_\otimes$ is full rank (which requires $\bar{P}$ to be full rank).

About the monotonicity
While the book explains this passage using the monotonicity of the exponential, this is nothing else than putting into play the equivalence between ML and MD criteria when we have AWGN noise.

Through the derivation which can be found from A signal processing perspective (pp. 57-130), page 57 to A signal processing perspective (pp. 57-130), page 58, we find that through the linear estimator we found we'll have

$$E^{\text{LS}} = \frac{N_0}{G}(\bar{P}_\otimes^* \bar{P}_\otimes)^{-1} \quad (2.239)$$

$$\hat{H}^{\text{LS}} = \frac{1}{\sqrt{G}}\bar{Y}\bar{P}^*(\bar{P}\bar{P}^*)^{-1}, \quad (2.247)$$

where (2.239) expresses the Mean-Square Error Matrix while (2.247) expresses the best estimation of the channel. The advantage of this equation with respect to (2.233) is that it entails less operations and also that since $\bar{P}$ is known, we can consider $\bar{P}^*(\bar{P}\bar{P}^*)^{-1}$ as already known.

Hmmmm

I'm sure there are more complicated and accurate reasons behind the procedure which gets us to (2.247), but for the time being I'd just observe that it seems as if we just switch $\bar{P}_\otimes$ and y and then replace $\bar{P}_\otimes \to \bar{P}^*$

A signal processing perspective (pp. 57-130), page 58

> Also notice that the ith row of $\bar{H}$ corresponds to the channel coefficients between the N_t transmit and the ith receive antenna, while the ith row of $\bar{Y}$ corresponds to the observation at the ith receive antenna

Due to how we defined $\bar{H}$, it doesn't look like this is true...we have that the ith row will contain coefficients such as $h_{00}[0], h_{01}[0], h_{00}[1], h_{01}[1]$, aka the coefficients of transmission from antenna 0 to all other antennas.

Skip

From page A signal processing perspective (pp. 57-130), page 58 to page A signal processing perspective (pp. 57-130), page 59 we find a brief discussion of sensible values for the pilots.
We can find the discussion of a LMMSE estimator at A signal processing perspective (pp. 57-130), page 61.

ML is one of the criteria! I know I received the joint probability distribution of all elements in the vector. What we want to do is to find the minimum of the probability density distribution while ***playing*** with our values. This is a pretty well known problem and has the solution we see in the slides. If the noise is Gaussian, at the end of the day we forget everything and minimize the distance we see. MAP requires to know the distribution of the input; we know that **if the input are equiprobable and independent, we'll have MAP $\equiv$ ML. If the noise is Gaussian, we have ML $\equiv$ MD.** This is why we're using the MD criterium. If we're doing minimum distance we can use LS and the solution is the one in the slides.

It's invertible if it's tall and we chose the pilots in an appropriate manner. There are indeed recipes to select the pilots; we should remember the line of reasoning; it **should not be taken for granted that stuff is invertible**. If we have to estimate at least four numbers and we're in the OFDM setting and we have $K = 256$, we need to make 256 estimates; the minimum is that we only need 8 numbers of $H[\text{numbers}]$.

The numbers in time domain are much less! We don't need to have 256 pilots, but we can't have less than 8 pilots. With less than 8 known variables we cannot estimate the 8 numbers in the matrices.

How do we distribute the pilots? We're going to distribute them evenly, otherwise we won't be able to get all the numbers.

In class

Derivation to simplify the LS estimator:

- Exploiting the fact that

$$\begin{aligned}(\bar{P}_{\otimes}^{*}\bar{P}_{\otimes})^{-1}\bar{P}_{\otimes}^{*} &= ((\bar{P}^{*}\otimes I)^{*}(\bar{P}^{T}\otimes I))^{-1}(\bar{P}^{*}\otimes I)^{*} \\ &= ((P\otimes I)(\bar{P}^{T}\otimes I))^{-1}(\bar{P}\otimes I) \\ &= ((P\bar{P}^{T})\otimes I)^{-1}(\bar{P}\otimes I) \\ &= (P\bar{P}^{T})^{-1}\bar{P}\otimes I,\end{aligned}$$

- Gives

$$\mathrm{vec}(\hat{H}) = \frac{1}{\sqrt{G}}((\bar{P}\bar{P}^{*})^{-1}\bar{P}^{*}\otimes I)\mathrm{vec}(\bar{Y})$$

- Exploiting $\mathrm{vec}(ABC) = (C^{T}\otimes A)\mathrm{vec}(B)$ in reverse

$$\begin{aligned}\hat{H} &= \frac{1}{\sqrt{G}}\bar{Y}((\bar{P}^{*}\bar{P}^{T})^{-1}\bar{P}^{*})^{T} \\ &= \frac{1}{\sqrt{G}}\bar{Y}\bar{P}^{*}(\bar{P}\bar{P}^{*})^{-1},\end{aligned}$$

We're not required to remember all of this, but it's recommended to go and check that.

Suppose we made all of our calculations and we have $\hat{H}$ and we're using OFDM; we know OFDM works with MIMO. If we know that the prefix is L, it means that we won't have more than L numbers in the input's response. A brutal technique is: suppose we are in the SISO setting and we have the $\hat{h}$ with 256 carriers. If $L = 16$, what should I expect? We should expect that the vector $\hat{H}$ should be nonzero from the element 0 to element 15.

From this we do IDFT and we have 256 elements:

$$h = \begin{bmatrix}\text{stuff}_0 \\ \vdots \\ \text{stuff}_{15} \\ 0 \\ \vdots \\ 0\end{bmatrix} \updownarrow 256$$

If we don't have zero, it means we have noise; we then just replace everything with zeros and we to the DFT: we have H^{*}, a better version of $\hat{H}$.

Assuming we haven't made a wrong decision on L (we thus must have $L < K$), we can cut everything after L, which will be noise. This is what we mean by saying ***we can work in the time domain***.

$$\widehat{H} = \begin{bmatrix} \\ \\ \end{bmatrix} \updownarrow 256 \xrightarrow[\text{IDFT}]{} \hat{h} = \begin{bmatrix} \updownarrow L \\ \rightsquigarrow \\ \text{noise} \end{bmatrix} \xrightarrow{\text{cut}} \begin{bmatrix} \updownarrow L \\ 0 \\ \vdots \\ 0 \end{bmatrix} \xrightarrow[\text{DFT}]{} H^*$$

The figure is showing that we can go in the time domain, cut everything after L (since it's noise) and then go back to the frequency domain, having a better signal.

9.3 Zero forcing equalizer for MIMO OFDM

We'd like to somehow adapt

> to the observation (we have $^{-1}$ and not † since we know that $\bar{P}_{\otimes}^{*}\bar{P}_{\otimes}$ is a square matrix).

so to estimate an OFDM transmission.

We can indeed extend what we saw to OFDM, considering that our objective is to distribute the various pilots in an even manner across the K subcarriers (A signal processing perspective (pp. 57-130), page 65), also considering that now our channel is defined as

$$\begin{aligned} \mathsf{y}[k] &= \sqrt{G}\mathsf{H}[k]\mathsf{p}[k] + \mathsf{v}[k] && (2.288)\\ &= \sqrt{G}\left(\sum_{\ell=0}^{L} H[\ell]e^{-j\frac{2\pi}{K}\ell k}\right)\mathsf{p}[k] + \mathsf{v}[k] \quad k = k_0, \dots, k_{N_p-1}, && (2.289)\end{aligned}$$

which, in terms of $\bar{H} = [H[0] \dots H[L]]$, gives

$$\begin{aligned} \mathsf{y}[k] &= \sqrt{G}\bar{H}\begin{bmatrix} I_{N_t} \\ e^{-j\frac{2\pi}{K}k} I_{N_t} \\ \vdots \\ e^{-j\frac{2\pi}{K}kL} I_{N_t} \end{bmatrix}\mathsf{p}[k] + \mathsf{v}[k] && (2.290)\\ &= \sqrt{G}\bar{H}\begin{bmatrix} \mathsf{p}[k] \\ e^{-j\frac{2\pi}{K}k}\mathsf{p}[k] \\ \vdots \\ e^{-j\frac{2\pi}{K}kL}\mathsf{p}[k] \end{bmatrix} + \mathsf{v}[k]. && (2.291)\end{aligned}$$

By defining

$$\mathsf{u}[k] = \begin{bmatrix} 1 & e^{-j\frac{2\pi}{K}k} & e^{-j\frac{2\pi}{K}k2} & \dots & e^{-j\frac{2\pi}{K}kL} \end{bmatrix}^T \tag{2.292}$$

9.4 A signal processing perspective

and through the derivation at A signal processing perspective (pp. 57-130), page 65 we see that

$$\mathsf{y}[k] = \sqrt{G}(\mathsf{u}[k] \otimes \mathsf{p}[k]) + \mathsf{v}[k] \tag{2.293}$$

and we eventually get to

$$\underbrace{\begin{bmatrix} \mathsf{y}[k_0] \\ \mathsf{y}[k_1] \\ \vdots \\ \mathsf{y}[k_{N_p-1}] \end{bmatrix}}_{\bar{\mathsf{y}}} = \sqrt{G} \underbrace{\begin{bmatrix} \mathsf{u}[k_0]^T \otimes \mathsf{p}[k_0]^T \otimes \mathsf{I}_{N_r} \\ \mathsf{u}[k_1]^T \otimes \mathsf{p}[k_1]^T \otimes \mathsf{I}_{N_r} \\ \vdots \\ \mathsf{u}[k_{N_p-1}]^T \otimes \mathsf{p}[k_{N_p-1}]^T \otimes \mathsf{I}_{N_r} \end{bmatrix}}_{\bar{\mathsf{P}}_\otimes} \text{vec}(\bar{H}) + \underbrace{\begin{bmatrix} \mathsf{v}[k_0] \\ \mathsf{v}[k_1] \\ \vdots \\ \mathsf{v}[k_{N_p-1}] \end{bmatrix}}_{\bar{\mathsf{v}}} \tag{2.296}$$

We need to ensure that $\bar{\mathsf{P}}_\otimes$ is tall, which requires

$$N_p \geq (L+1)N_t$$

Elaborating a lil bit...

- u is $(L+1) \times 1$, p is $N_t \times 1$;
- $\mathsf{u} \otimes \mathsf{p}$ is thus $(L+1)N_t \times 1$;
- Then we must have that $\mathsf{u}^T \otimes \mathsf{p}^T \otimes \mathsf{I}_{N_r}$ is $N_r \times N_r N_t(L+1)$;
- Finally, we stack a bunch of those (N_p) on top of one another, getting $N_r N_p \times N_r N_t(L+1)$.

We can now see that

$$N_r N_p \geq N_r N_t(L+1) \implies N_p \geq N_t(L+1).$$

The number of pilots is equal to the one of frequency flat channels and is lower of L when compared with frequency selective channels.

We could also have frequency domain estimation, see brief discussion at A signal processing perspective (pp. 57-130), page 66.

Examples (2.24, 2.30, 2.31)

Example 2.24: Least-square estimator for a frequency-flat SISO channel

Normally our estimator is

Through the derivation which can be found from A signal processing perspective (pp. 57-130), page 57 to page 58, we find that through the linear estimator we found we'll have

$$E^{LS} = \frac{N_0}{G}(\bar{P}_\otimes^* \bar{P}_\otimes)^{-1} \tag{2.239}$$

$$\hat{H}^{LS} = \frac{1}{\sqrt{G}} \bar{Y}\bar{P}^*(\bar{P}\bar{P}^*)^{-1}, \tag{2.247}$$

where (2.239) expresses the Mean-Square Error Matrix while (2.247) expresses the best estimation of the channel.

The advantage of this equation with respect to (2.233) is that it entails less operations and also that since $\bar{P}$ is known, we can consider $\bar{P}^*(\bar{P}\bar{P}^*)^{-1}$ as already known.

Here though we have both that $L = 0$ (meaning that $\bar{H} \to H$) and that $N_t = N_r = 1$, meaning that we're in a SISO case, such that $H \to h$ and that our $\bar{P}$ matrix will reduce to a horizontal vector containing input pilots for $p[0], \dots, p[N_p - 1]$.

In order to have

Due to the discussion we made about zero forcing equalizers, **we need to ensure that $\bar{\mathsf{P}}_\otimes$ is square or tall**, requiring to check that

$$N_p \geq (L+1)N_t + L \tag{2.234}$$

and also that $\bar{\mathsf{P}}_\otimes$ is full rank (which requires $\bar{P}$ to be full rank).

To be satisfied, we require that $N_p \geq 1$ so we just choose $N_p = 1$, implying that $\bar{P} \to p[0] = \sqrt{E_s}e^{j\phi}$.

The rest follows quite easily:

Example 2.24 (Least-squares estimator for a frequency-flat SISO channel)

Let $L = 0$ and $N_t = N_r = 1$. For this simplest of settings, a single scalar pilot suffices. With such a pilot being $\sqrt{E_s}e^{j\phi}$ where ϕ is an arbitrary phase, (2.247) reduces to

$$\hat{h} = \frac{e^{-j\phi}}{\sqrt{G}E_s} y \tag{2.249}$$

while E in (2.239) reduces to

$$\begin{aligned} \mathsf{MMSE} &= \frac{N_0}{GE_s} && (2.250) \\ &= \frac{1}{\mathsf{SNR}}, && (2.251) \end{aligned}$$

which does not depend on ϕ; the pilot's phase can thus be set to zero.

For $N_p > 1$, $\bar{P}$ becomes a row vector and the pilot sequence requires no special structure: in this setting, $\bar{P}\bar{P}^*$ is always a scalar, even if the same pilot symbol is simply repeated. It is easy to verify that

$$\mathsf{MMSE} = \frac{1}{N_p \mathsf{SNR}}, \tag{2.252}$$

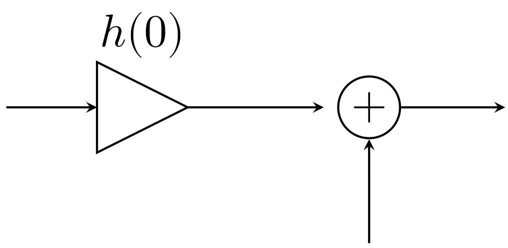
indicating that the accrual of energy over multiple pilots has the same effect as increasing the energy of a single pilot.

Professor's comments (Hz 616 ex 2.29)

We have $h[\ell] \neq 0$ only for $\ell = 0$: this is what ***frequency flat*** means.

Approximation Logic: Consider the term $\frac{1}{1+x}$:

- For $x \gg 0 \implies \approx \frac{1}{x}$
- For $x > 0$ (small) $\implies \approx 1 - x$



We only need $h(0)$ in order to have an estimate of the channel. We just take the output and we divide it by what we had transmitted. The error we get is 1 over the SNR. If the noise goes to zero, we have infinite error and infinite SNR$\rightarrow$no error.

What we can do is to send several symbols one after the other; of course we're paying in the fact that we're sending multiple symbols.

If we use N_p pilots, the power of the noise is divided by N_p: we can gain much in this sense but of course nothing comes for free and while we get this, we're also not sending information: we loose in the bitrate but we gain in the accuracy detection.

Example 2.30 (LMMSE estimator for a vec channel)

Here we're instead asked to find the LMMSE estimator, which we know to be defined as
The estimator is simply

$$W^{\text{MMSE}} = R_y^{-1} R_{ys}. \tag{1.224}$$

; since we have IID and unit-vector values in $\bar{H}$, we have that

$$\begin{aligned} R_{\text{vec}(Y)} &= \mathbb{E}[G\bar{P}_{\otimes}\text{vec}(\bar{H})\text{vec}^*(\bar{H})\bar{P}_{\otimes}^* + \underbrace{0+0}_{\text{Indep.}} + \text{vec}(\bar{H})\text{vec}^*(\bar{H})] \\ &= G\bar{P}_{\otimes}\bar{P}_{\otimes}^* + N_0 I \end{aligned}$$

and also that

$$R_{\text{vec}(\bar{Y})\text{vec}(\bar{H})} = \mathbb{E}[\sqrt{G}\bar{P}_{\otimes}\text{vec}(\bar{H})\text{vec}^*(\bar{H})] = \sqrt{G}\bar{P}_{\otimes},$$

in which btw we used $\bar{H}$ instead of the supposed signal because here we're trying to estimate $\bar{H}$, not the channel. From this, it follows that

Example 2.30 (LMMSE estimation for an IID MIMO channel)

Let the entries of $\bar{H}$ be IID and, by normalization, of unit variance. Then, $R_{\text{vec}(H)} = I$ and

$$W^{\text{MMSE}} = \sqrt{G}\bar{P}_{\otimes}(N_0 + G\bar{P}_{\otimes}^{*}\bar{P}_{\otimes})^{-1}, \tag{2.270}$$

from which

$$\text{vec}(\hat{H}) = \sqrt{G}\left(G\bar{P}_{\otimes}^{*}\bar{P}_{\otimes} + N_0 I_{(L+1)N_t N_r}\right)^{-1}\bar{P}_{\otimes}^{*}\text{vec}(\bar{Y}), \tag{2.271}$$

while the MMSE matrix in (2.262) specializes to

$$E = I - G\bar{P}_{\otimes}^{*}(G\bar{P}_{\otimes}\bar{P}_{\otimes}^{*} + N_0 I)^{-1}\bar{P}_{\otimes}. \tag{2.272}$$

In the foregoing expression for $\text{vec}(\hat{H})$, the dimensionality of the identity matrix has been noted explicitly to set the stage for the decomposition $N_0 I_{(L+1)N_t N_r} = N_0 I_{(L+1)N_t} \otimes I_{N_r}$. Recalling that $\bar{P}_{\otimes} = \bar{P}^T \otimes I_{N_r}$, we can then write

$$\begin{aligned}
\text{vec}(\hat{H}) &= \sqrt{G}\left(G\bar{P}^c\bar{P}^T \otimes I_{N_r} + N_0 I_{(L+1)N_t} \otimes I_{N_r}\right)^{-1}(\bar{P}^c \otimes I_{N_r})\text{vec}(\bar{Y}) && (2.273)\\
&= \left[\sqrt{G}(G\bar{P}^c\bar{P}^T + N_0 I_{(L+1)N_t})^{-1}\bar{P}^c \otimes I_{N_r}\right]\text{vec}(\bar{Y}) && (2.274)
\end{aligned}$$

and, resorting once more to the identity $\text{vec}(ABC) = (C^T \otimes A)\text{vec}(B)$, what ensues is

$$\hat{\bar{H}} = \sqrt{G}\bar{Y}\bar{P}^{*}(G\bar{P}\bar{P}^{*} + N_0 I)^{-1}, \tag{2.275}$$

which coincides with its least-squares counterpart in (2.247), save for the argument of the inverse being regularized in proportion to the noise strength. This similarity is consistent with the fact that the channel statistics, which differentiate the least-squares and LMMSE estimators, are trivial when the entries of $\bar{H}$ are IID; an SNR-dependent regularization suffices to optimally bias the estimate of each realization. The possibility of unvectorizing the estimate in the form (2.275) is also a consequence of the receive antennas being uncorrelated, as then the LMMSE estimate of each row of $\bar{H}$ can be produced separately, based only on the corresponding row of $\bar{Y}$.

Professor's comments.

Entries are random: we're dealing with a stochastic channel! We want to estimate this channel. If we want to estimate a random variable, we want to estimate the probability density function of the random variable or some other complete description of that variable. We expect the channel is random but we need to deal with the realization.

Realization

If I have a Bernoulli, before throwing the coin we have a random variable; when we're looking at the result, we have the realization of that random variable. If we take many realization we expect to have half heads and half tails (This of course is the frequentistic approach). We can only estimate a realization of random variable!

We have the realization, we do the calculation and again we have a noise term to the minus one. It's what is called a regularization term, which ensures the matrix is always invertible (refer to pag 46 slide).

There's a nice example.

We have

$$\begin{bmatrix} 1 & -1 \\ \rho & -1 \end{bmatrix}$$

for which we see what's up when ρ changes between -1 and 1. **The matrix will become singular when $\rho \to 1$. If we use the same noise and we have $\rho = 1$, we'll see that the noise of zero forcing will escalate while the MS will stay still** (the zero forcing tries to invert a singular matrix!).

> [color=ForestGreen!50!black] Random: we need to check for consistency when we look at our results.